

Exercise sheet 2

Data – values, variables, data types and expressions

Not graded, not handed in. Do them anyway. This is where the learning happens! Solutions are published after we discuss them in class.

Predict before you run. Write the answer down first, then check it in Python. On the exam there will be no Python, so the paper answer is the one that counts.

Part A — Values, variables and types

A1 Value and type

For each expression, give **both** the value and the type (int, float, str or bool).

expression	expression
1 <code>7 + 3</code>	7 <code>"7" + "3"</code>
2 <code>7 / 1</code>	8 <code>3 * "ab"</code>
3 <code>2 ** 3 ** 2</code>	9 <code>10 % 3 == 1</code>
4 <code>7 // 2 + 7 % 2</code>	10 <code>int("7") + 3</code>
5 <code>-9 // 4</code>	11 <code>float(7)</code>
6 <code>8 / 4 * 2</code>	12 <code>int(-3.9)</code>

Two of these are not what most people guess. When we go through them, mark which two caught you.

A2 Trace it

```
a = 3
b = 8
a = a + b
b = b - a
a = a * 2
b = a % b
```

- Draw a table with one column for a, one for b, and one row per line. Fill it in.
- What are the final values?
- Line 6 would fail if b were 0 at that point. What error would Python raise? (Guess the name — you have met two of the three error names already.)

A3 *Legal, and good*

For each name, say whether it is (i) legal Python, and (ii) a good choice. The two questions have different answers.

total_price	2ndAttempt	x	TotalPrice
class	price_in_euros	thing	n_students
Price	my variable	1	temperature_c

One of the legal ones is a bad idea for a reason that has nothing to do with style. Which, and why?

A4 *Brackets in the wrong place*

Each line is meant to compute the thing described in the comment, and each gets it wrong. Fix the brackets.

```
# the average of a, b and c
average = a + b + c / 3

# the area of a circle of radius r, with PI already defined
area = PI * r * 2

# one half of the sum of x and y
half = 1 / 2 * x + y

# the total cost of n items at price p, with 22% VAT added
total = n * p * 1 + 0.22
```

For each one, also say what the broken version actually computes.

A5 *Negative numbers*

- Work out $-7 // 2$ and $-7 \% 2$ by hand.
- Check them in Python. Were you right?
- $//$ does not round towards zero, it rounds *down*. Explain in one sentence why that makes $-7 // 2$ equal to -4 .
- Given that $a // b$ and $a \% b$ must satisfy $(a // b) * b + (a \% b) == a$, work out what $-7 \% 2$ has to be.

A6 *The one from last week*

Last week's sheet ended with a change calculation that printed `12.799999999999997`, and we said the explanation would have to wait.

- In one or two sentences, explain it now.
- Write a line of Python that tests whether $0.1 + 0.2$ and 0.3 are equal *in a way that gives True*.
- Does the same problem affect `int`? Why, or why not?

Part B — Writing small programs

For every question in this part, **write the comment plan first**, then fill in the code underneath it. If you cannot write the comments, you do not yet understand the problem, and typing will not help.

B1 *A circle*

Given a radius of 3.5 and $PI = 3.14159$, print the area and the circumference, each labelled.

Then change the radius to 1 and run it again. You should have to edit exactly one line. If you had to edit more, your program has the same number written in two places — fix that.

B2 *A shopping total*

You buy 3 notebooks at €2.40 each and 2 pens at €1.15 each. VAT of 22% is added to the total.

Print the total before VAT, the VAT itself, and the total to pay.

B3 *Seconds, again*

Write a program that turns `total_seconds = 10000` into hours, minutes and seconds, and prints them.

Use only `//` and `%`. Work out the plan on paper before writing any code — it is four lines, but only if you get the order right.

B4 *Swap*

`a = 12` and `b = 5`. Swap their values so that `a` becomes 5 and `b` becomes 12, **without** introducing a third variable.

Then do it a second way, *with* a third variable. Which version would you rather read in six months?

B5 *What does it print?*

Predict the output of each, then check.

```
# 1
x = 5
y = x
x = 10
print(y)

# 2
n = 7
n += 3
n *= 2
print(n)

# 3
print(10 / 2)

# 4
price = "20"
print(price + "5")

# 5
a = 1
```

```
b = 2
print(a == b, a < b, a + b == 3)
```

Number 1 is the one to think about carefully. Number 4 is legal Python and almost certainly not what its author wanted.

B6 *Three broken programs*

Name the error each one raises, and fix it.

```
# 1
age = "24"
next_year = age + 1
print(next_year)

# 2
total = 0
total = total + count
print(total)

# 3
n = int("seven")
print(n * 2)
```

B7 *It runs, and it is wrong*

This program is supposed to print the average of three marks. It runs without any error, and prints the wrong number.

```
mark1 = 28
mark2 = 30
mark3 = 25
average = mark1 + mark2 + mark3 / 3
print("average:", average)
```

- a) What does it print?
- b) What should it print?
- c) Fix it.
- d) This is the most dangerous kind of bug. Say in one sentence why.

If you finish early

Take your program from **B2** and make it handle a different VAT rate by changing one line only. Then add a second product and see how much of the program you had to touch.

Keep the answer in mind: next week we will meet the tools that stop this getting worse.